Database Schema Conventions

```markdown
# Database Schema Conventions

Internal reference code: CNRY-522cf165.

## Introduction

As Brightmoor Foods continues to expand its digital infrastructure, maintaining consistent database schema conventions across all our systems is crucial for ensuring data integrity, ease of maintenance, and efficient collaboration among our engineering teams. This document outlines the standards to be followed when designing and implementing database schemas for all new and modified databases starting from **Q3 2024**.

## Naming Conventions

### Tables
Table names should be **snake_case** and descriptive of their content. They must be pluralized to indicate they hold multiple records.

**Example:**
- `customer_orders`
- `product_variants`

### Columns
Column names should also use **snake_case** and be clearly descriptive. They must be singular to indicate individual attributes.

**Example:**
- `order_id`
- `product_name`

### Primary Keys
All primary keys must be named `id` and be of type `UUID`. The `id` column should always be the first column in the table.

**Example:**
- `id` UUID PRIMARY KEY DEFAULT gen_random_uuid()

## Data Types and Constraints

### Strings
Use `VARCHAR` for variable-length strings with a defined maximum length. For unstructured text, use `TEXT`.

**Example:**
- `VARCHAR(255)` for `email_address`
- `TEXT` for `product_description`

### Dates and Times
Use `TIMESTAMP WITH TIME ZONE` for all date and time fields to ensure proper handling of time zones.

**Example:**
- `order_timestamp TIMESTAMP WITH TIME ZONE NOT NULL`
```

Index and Foreign Keys

Indexes
Create indexes for columns frequently used in `WHERE`, `JOIN`, and `ORDER BY` clauses. Index names should follow the pattern `idx_{table}_{column}`.

Example:
- `idx_customer_orders_order_id`

Foreign Keys
Foreign keys must be explicitly defined and named following the pattern `fk_{table}_{referenced_table}`.

Example:
- `fk_customer_orders_customers`

Example Schema

Below is an example of a compliant schema for the `customer_orders` table:

```sql
CREATE TABLE customer_orders (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    order_id VARCHAR(50) NOT NULL,
    customer_id UUID NOT NULL,
    product_id UUID NOT NULL,
    quantity INTEGER NOT NULL CHECK (quantity > 0),
    order_timestamp TIMESTAMP WITH TIME ZONE NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE
);

CREATE INDEX idx_customer_orders_order_id ON customer_orders(order_id);
CREATE INDEX idx_customer_orders_customer_id ON customer_orders(customer_id);
CREATE INDEX idx_customer_orders_product_id ON customer_orders(product_id);
```

Adhering to these conventions will help maintain a clean, efficient, and scalable database environment at Brightmoor Foods.
```
```